ALL

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3. Question 3

The malloc() function in C/C++ allocates memory at which time/location?

Pick **ONE** option

- compilation time on stack
- Linking time on stack
- Load time on heap
- Execution time on heap
- Execution time on stack

Clear Selection

Answered: 0 / 17 01 hour 11 mins

ALL

2. Question 2

In a company, 26 employees are in the HR department, 32 are in the technical department, and 30 are in the management department. Of these, 5 employees are in all three departments, while 37 are in only one of them. How many employees like only two of the three departments?

Pick ONE option

- 36
- 18
- 88
- 27

Clear Selection

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ALL

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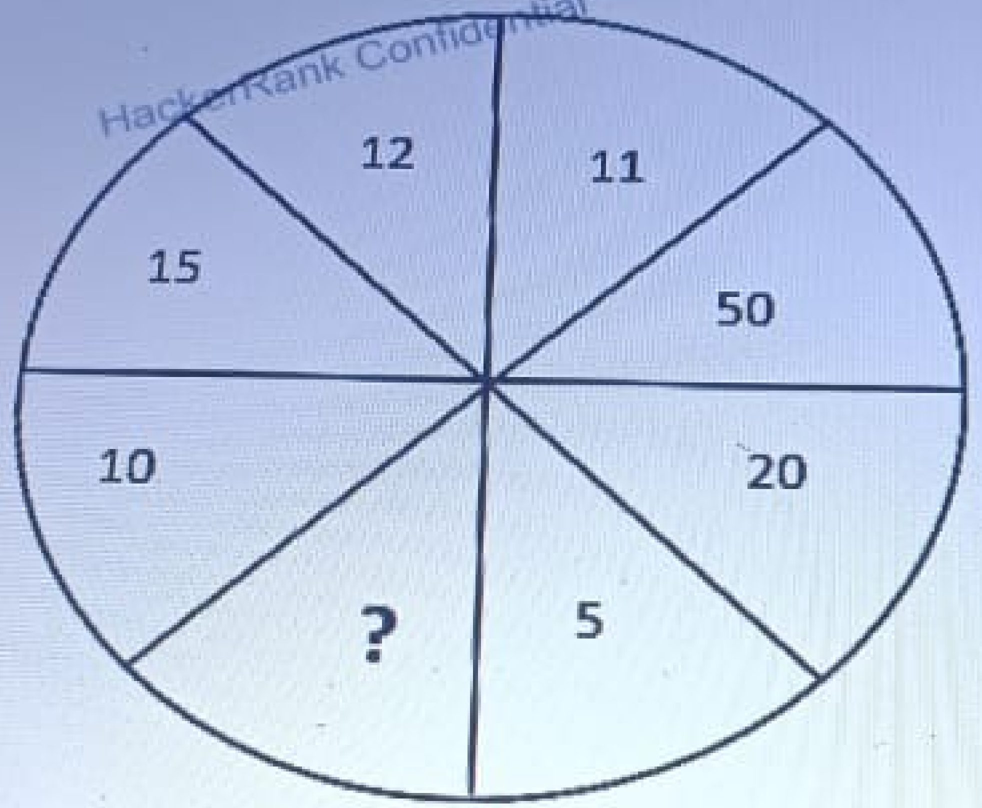
5. Question 5

What number logically replaces (?).



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DELL

ALL

4. Question 4

A circular linked list can be used to implement

Pick **ONE** option

- A stack
- A queue
- Both
- Neither

Clear Selection

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ALL

6. Question 6

Which one of the following state is FALSE?

A) Three packets are sent to establish TCP connection
B) The ACK packet from receiver during TCP connection initiation has both SYN and ACK bit set
C) Four packets are sent to terminate the TCP connection

Pick **ONE** option

- Only A
- Only C
- Both B and C
- None to these

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8. Question 8

A Linux system administrator needs to securely transfer a file named *myapp.log* from a local server to a remote server using SSH. The sys
user privileges, and the *myapp.log* file can only be accessed by the root user.

Which command should the administrator use?

Pick **ONE** option

```
sudo scp -i /root/.ssh/id_rsa /path/to/myapp.log ubuntu@10.0.1.34:/remote-dir/my-location/
```

```
sudo -u root scp /path/to/myapp.log ubuntu@10.0.1.34:/remote-dir/my-location/
```


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7. Question 7

Select the data structure suitable for implementing a dictionary with key-value

Pick **ONE** option

- Array
- Stack
- Hash Table
- Queue

Clear Selection

9. Question 9

What is the maximum number of keys in a B-tree of order 6 and of height 4?

In computer science, a B-tree is a self-balancing tree data structure that maintains sorted data and allows searches, sequential access, insertions, and deletions in logarithmic time. B Tree is **a specialized m-way tree** that is widely used for disk access.

Pick **ONE** option

- 7776
- 7775
- 16384
- 16383

Clear Selection

11. Question 11

Consider the following recurrence relation:

$$R(n) = R(n-1) + 1/n$$

Find the time complexity of R(n)

Pick **ONE** option

- O(log n)
- O(n)
- O(n log n)
- O(log log n)

Clear Selection

13. Question 13

Guess the output for the following program:

```
public static int fun(int x, int y) {
    if (n == 0)
        return 1;
    else if (n % 2 == 0)
        return fun(x * x, n / 2);
    else
        return x * fun(x * x, (n - 1) / 2);
}
public static void main() {
    int ans = fun(2, 10);
    System.out.println(ans);
}
```


Pick **ONE** option

- 1023

15. Question 15

Consider the following IP addresses:

A. 201.119.1.30
B. 201.119.1.33
C. 201.119.1.46

If subnet mask is 255.255.255.240 then which IP belong to same subnet.

Pick **ONE** option

- A and B
- B and C
- A, B and C
- None

Clear Selection

Answered: 0 / 17 01 hour 11 mins

ALL

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Pick ONE option

- 36
- 18
- 88
- 27

Clear Selection

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4. Question 4

Which file contains the default environment variables when using the bash shell?

Pick ONE option

- ~/.profile
- ~/.bash
- /etc/profile.d
- ~/bash

Clear Selection

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5. Question 5

The malloc() function in C/C++ allocates memory at which time/location

Pick ONE option

- compilation time on stack

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- O(log n)
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Clear Selection

ALL

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Pick **ONE** option

- 7776
- 7775
- 16384
- 16383

Clear Selection

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- Linking time on stack
- Load time on heap
- Execution time on heap
- Execution time on stack

Clear Selection

BETA Can't read the text? Switch theme

6. Question 6

A Linux system administrator responsible for process monitoring and manage

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7. Question 7

Which of the following commands will list the last three commands you run excluding itself with the history ID of these commands:

Pick ONE option

- history -n 3
- history -c 3
- fc -l -3
- fc -ln -3

Clear Selection

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8. Question 8

Which one of the following statements is correct?

Pick ONE option

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8. Question 8

Which one of the following statements is correct?

Pick ONE option

- A macro must be defined in capital letters.
- Once preprocessing is over and the program is sent for the compilation, the macros are removed from the expan… source code.
- Macros have a local scope.
- In a macro call, the control is passed to the macro.

Clear Selection

BETA Can't read the text? Switch theme

9. Question 9

Consider a complex multi-threaded Linux application that utilizes POSIX threads for parallel pr… protected by a mutex to ensure proper synchronization and prevent data corruption. Howev…

1. The thread that holds the mutex enters a non-preemptive state and cannot be forcibly pree…

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16. Question 16

In order to uniquely identify the devices on the computer network, each device is allotted an Internet Protocol address (IP address). There are 2 versions of address formats in use i.e. IPv4 which 32 bits in length and IPv6 having 128 bits. Modern devices use IPv6 but in order to communicate with devices supporting only IPv4, the conversion from IPv4 to IPv6 and vice versa. Also, the IPv4 addresses of the format `127.x.x.x` (where x means any octet value) are the loopback addresses and the equivalent of it in IPv6 is `::1`.

Consider the IP address `192.168.10.92`. The 4 octets of the IP address would be 192, 168, 10 and 92 and the hexadecimal equivalent of these octets are C0, A8, 0A and 5C. Then, the first and second are concatenated and third and fourth are concatenated to get C0A8 and 0A5C. Finally, the IPv6 is formed as `::FFFF:C0A8:0A5C`. (The FFFF in the IPv6 address is a constant

```
/*
 * Complete the 'convertToIpv6' function below.
 *
 * The function accepts STRING ipv4address as parameter.
 */

void convertToIpv6(string ipv4address) {

}

int main() ...
```

Test Results

Custom Input

17. Question 17

Given a string animal_sequence that consists of animal names separated by the separator character '-', your task is to write a function that returns the length of the last animal name in the sequence. An animal name is defined as a sequence of characters that does not contain the separator character '-'. The function should ignore any trailing or leading characters in the animal_sequence

test_case_1 = "Elephant-Tiger-Lion-"
Output: 4
test_case_2 = "|Horse-Dog-Cat-|"
Output: 3

Language C++20

Autocomplete Ready

```
> #include <bits/stdc++.h>

/*
 * Complete the 'last_animal_name_length' function below.
 *
 * The function is expected to return an INTEGER.
 * The function accepts STRING animal_sequence as paramete
 */

int last_animal_name_length(string animal_sequence) {

}

> int main() ...
```

Test Results

Custom Input

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Clear Selection

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Pick ONE option

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BETA Can't read the text? Switch theme

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- Linking time on stack
- Load time on heap
- Execution time on heap
- Execution time on stack

Clear Selection

BETA Can't read the text? Switch theme

6. Question 6

A Linux system administrator responsible for process monitoring and manage

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Clear Selection

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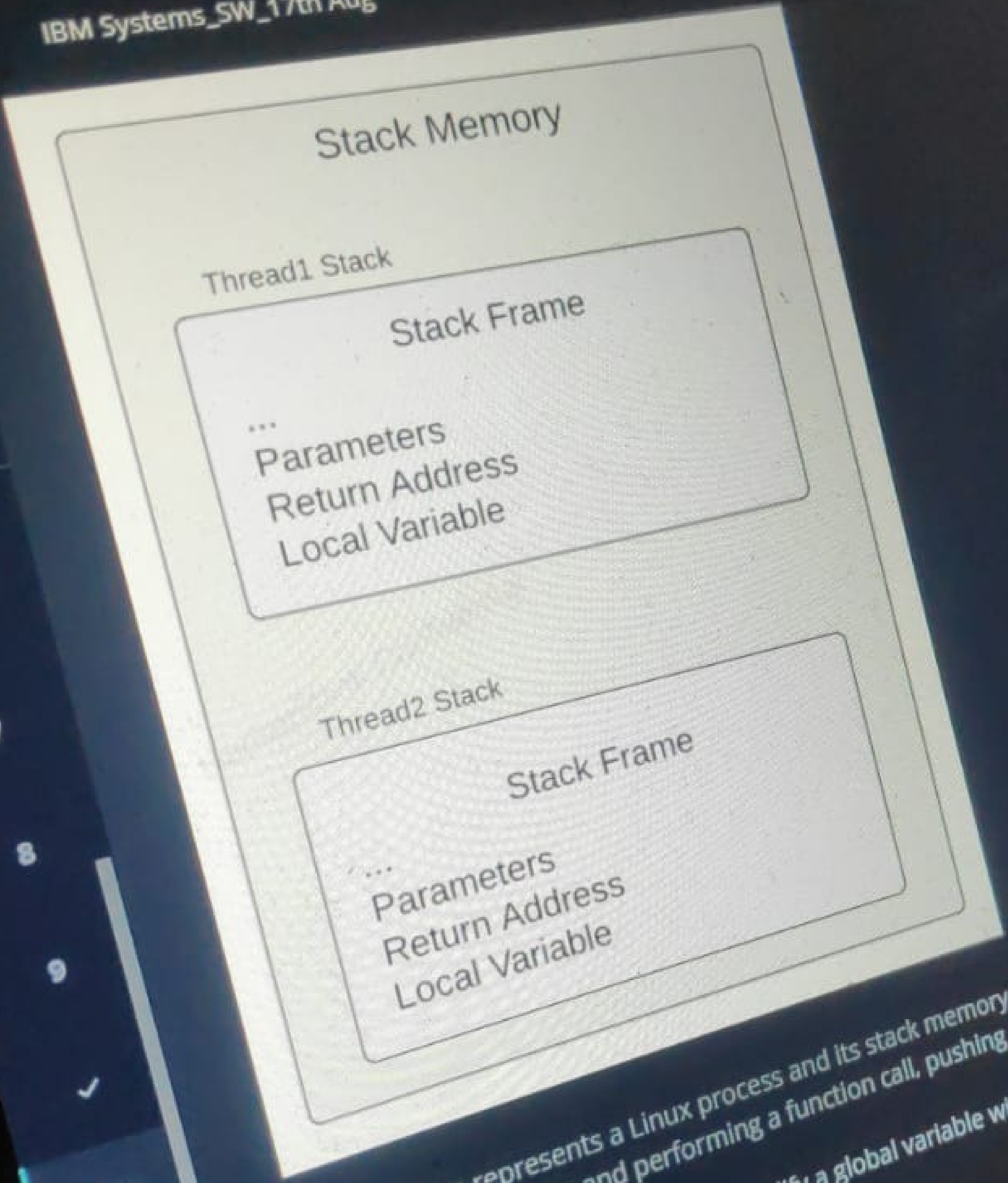

This diagram represents a Linux process and its stack memory layout. The process has two threads, Thread1 a
Thread 1 is executing and performing a function call, pushing data onto its stack. Simultaneously, Thread 2 is a

What will happen if both threads modify a global variable within the function?

Pick ONE option

- The global variable will be overwritten with the value from the thread that called the function last
- The global variable will be unaffected, as each thread has its own stack, isolating their modifica
- The final value will be unpredictable since the order of their executions cannot be determin

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12. Question 12

Number represent by y in the below equation

$(1101)_2 = (y)_3$

Pick **ONE** option

- 120
- 101
- 102
- 111

Clear Selection

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13. Question 13

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12
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15

How many times does "Hello" get printed?

```
int main() {
 printf("Hello");
 fork();
printf("Hello");
}
```

Pick **ONE** option

- 2
- 3
- 4

12. Question 12

Which file contains the default environment variables when using the bash shell?

Pick **ONE** option

- ~/.profile
- ~/.bash
- /etc/profile.d
- ~/bash

Clear Selection

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Answered: 12 / 17 23 mins

ALL

This diagram represents a Linux process and its stack memory layout. The process has two threads, Thread1 and Thread2, each with its own stack, sharing the same address space.
Thread 1 is executing and performing a function call, pushing data onto its stack. Simultaneously, Thread 2 is also executing and making its own function call, pushing data onto its stack. Both threads are using the same function with different arguments.

What will happen if both threads modify a global variable within the function?

Pick **ONE** option

- The global variable will be overwritten with the value from the thread that called the function last.
- The global variable will be unaffected, as each thread has its own stack, isolating their modifications.
- The final value will be unpredictable since the order of their executions cannot be determined.
- The global variable will automatically be protected, and only one thread can modify it at a time.

Clear Selection

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Answered: 14 / 17 20 mins

A Linux system administrator responsible for process monitoring and management must identify and terminate a process that is consuming excessive CPU resources. The process is identified by its command-line arguments, which include the string *app_process* and the value *myapp* as an argument.

Which command-line snippet accomplishes this?

Pick ONE OR MORE options

- [ ] `ps aux | grep -E "app_process.*myapp" | awk '{print $2}' | xargs kill`
- [ ] `pgrep -a "app_process myapp" | awk '{print $1}' | xargs kill`
- [ ] `pgrep -f "app_process.*myapp" | xargs kill`

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